‘Objectsville’

covering

** Procedural *versus* Object-Oriented programming

Chapters 2-4 – “Head First Java” book
Chapters 4+6 – “Introduction to Java Programming” book

Concepts: OO programming and objects

What is OO programming?

- Constructing software systems which are structured collections (or sets) of classes.
- These classes produce instances called objects, which communicate with each other using messages.

What is an object?

- An object is a thing; it is a fundamental entity in Java.
- Objects tend to be the nouns in specifications.
- In software terms, e.g.
  - car;
  - bank account;
  - student;
  - employee;
  - complex number;
  - GUI button.

Imagine that you are building a database about cars; you would have a Car object.

OO = Object-Oriented

What is not an object?

- Attributes (or states) of an object: essentially anything that describes or quantifies an object.
  - speed, colour, make, model, and position are all attributes of a car object;
  - number, owner, balance might be attributes of a bank account object.

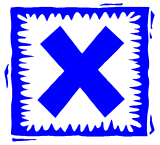
- Operations (or behaviours) of an object: they mostly correspond to verbs in a requirements specification.
  - turn left, speed up, slow down, turn right are all operations of a car object.
  - open, close, deposit, withdraw, are all operations on a bank account object.

What is a class?

- An object is defined by a class.
- The class defines the attributes and operations exposed by one or more related objects
- In Java, a class is to:
  - define a kind of object or in other words, to define a data type

Classes *versus* Objects
An object is an instance of a particular class.

- In the examples so far, all the code went in the `main()` method.



This is not object-oriented!

- How does object-oriented programming change how we do things?
- We can split up code between different objects …

… and things for you to try out!

Real world objects (1/3)

Person

To describe a person: name, gender, age, occupation, …
A person can do: eat, drink, sleep, walk, …

object

Jane
female
19
Student
…

object

Emma
female
45
Doctor
…

Car

To describe a car: make, model, year, colour, …
A car can do: accelerate, brake, turn, reverse, …

object

BMW
M3
blue
…
…

object

Mini
Cooper
red
…
…

Real world objects (2/3)

What are the attributes and operations of a mobile phone?

Mobile phone

Attributes:

Operations:

object

iPhone
5
Vodafone
…

object

Samsung
Galaxy S3
Orange
…

object

BlackBerry
Curve 8900
O2
…

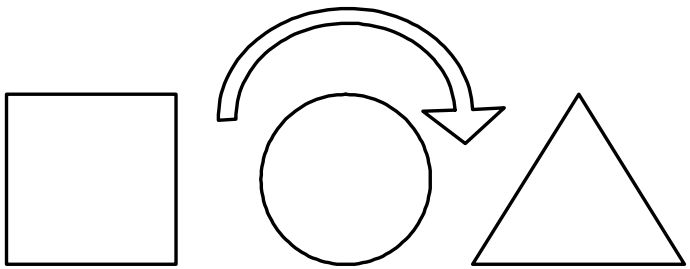
Real world objects (3/3)

Bank account

Attributes	Operations
account number	view details
account name	print statements
account type	check balance
sort code	deposit
address	withdraw
balance	change address
overdraft limit	change overdraft limit
…	…

OO *versus* Procedural Programming

The Specification
There will be shapes on a GUI: a square, a circle and a triangle. When the user clicks on a shape, the shape will rotate clockwise 360° (i.e. all the way around) and play an MP3 sound file specific to that particular shape.



- The task: to create a program that fullfils the following specification
  - Procedural approach
    - What does the program have to do?
    - What procedures are needed?
      - **rotate** and **playSound**

Procedural

```
rotate(shapeNum) {
  // make the shape rotate 360°
}
playSound(shapeNum) {
  // use shapeNum to loop-up which
  // sound to play and play it
}
```

OO: questions to answer (1/2)

- What are the things in this program, i.e. what are the objects?
  - Objects are things or nouns.
  - The main interacting force of this program is of course the shapes.

The Specification

There will be shapes on a GUI, a square, a circle and a triangle. When the user clicks on a shape, the shape will rotate clockwise 360° (i.e. all the way around) and play an MP3 sound file specific to that particular shape.

There are other objects, but this is enough to start with.

OO: questions to answer (2/2)

- Verbs indicate the actions of an object.

In Java, objects correspond to classes!

The Specification

There will be shapes on a GUI, a square, a circle and a triangle. When the user clicks on a shape, the shape will rotate clockwise 360° (i.e. all the way around) and play an MP3 sound file specific to that particular shape.

Object-Oriented

```
Square
rotate() {
  // code to rotat
}
playSound() {
  // code to play
  // sound for Squ
}
```

```
Triangle
rotate() {
  // code to rota
}
playSound() {
  // code to play
  // sound for Tr
}
```

```
Circle
rotate() {
  // code to rotate
}
playSound() {
  // code to play
  // sound for Circle
}
```

What if the specification changes …

Addition to Specification
A fourth shape is needed – a random shape. When the user clicks on the random shape, it will rotate and play a WAV sound file.

- `rotate()` will still work, as the code uses a lookup table to match a shape to a graphic.
- However, `playSound()` has to change!

Procedural

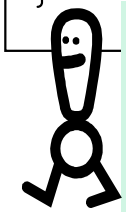
```
playSound(shapeNum) {
  // if shape is not a random shape
  // as before..
  // else
  // play random shape
}
```

Always try to minimise altering code you have already tested. Changes could introduce errors!

Object-Oriented

RandomShape

`rotate() {`
`  // code to rotate`
`}`
`playSound() {`
`  // code to play`
`  // sound for RandomShape`
`}`



With OO, we do not need to ‘touch’ any of the code we have already written!

Procedural and OO programming: again

- In order to account for variation in rotation points, we need to add some new arguments in the procedural method.

Procedural

```
rotate(shapeNum, xPt, yPt) {
  // if shape is not a random shape
  // calculate the centre point
  // based on a rectangle and rotate;
  // else use passed in xPt and yPt
  // as rotation offset and rotate
}
```

Lots of code has been affected! It will ALL have to be recompiled and tested.

new attributes

Object-Oriented

```
RandomShape
int xPoint
int yPoint
rotate() {
  // code to rotate
}
playSound() {
  // code to play
  // sound for RandomShape
}
```

New rules for rotation are simply put in the **`RandomShape`**'s **`rotate()`** method. No other shape is affected! The old shapes don't need to be tested again. In fact, the compiled code for **`Circle`**, **`Square`** and **`Triangle`** doesn't change at all.

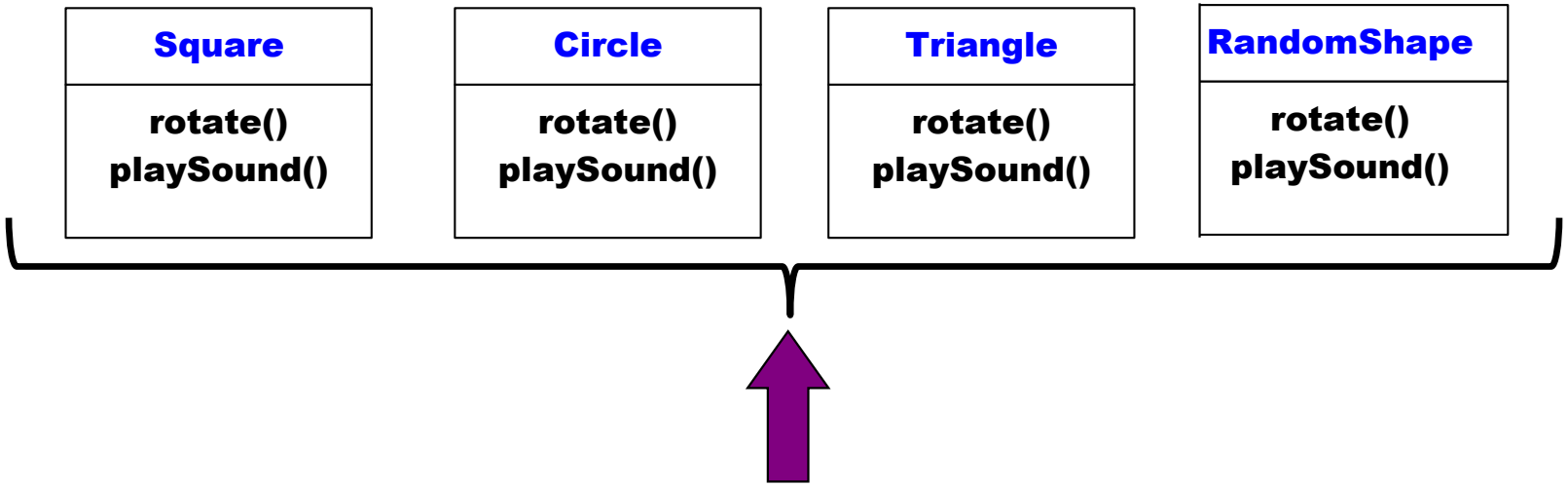
Summary: Procedural *versus* OO Approaches

- Procedural approach
  - 2 procedures that behave differently based on the shape.
  - If anything about the specification changes, then these 2 methods have to change.

- OO approach
  - 4 classes, with 2 methods each!
    - Many more methods than with the procedural approach!
    - Duplicated code? ← Is this a good thing?
  - However, each object controls its own behaviour.

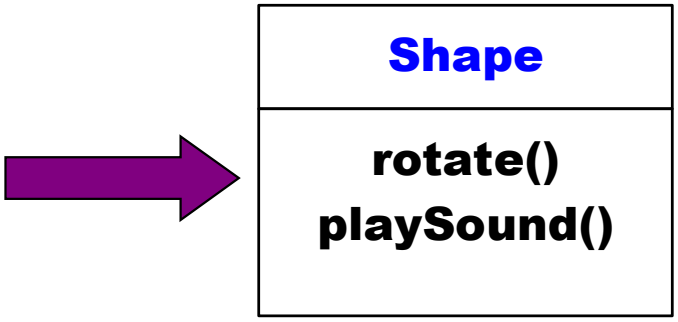
… and things for you to try out!

Introducing Abstraction and Inheritance

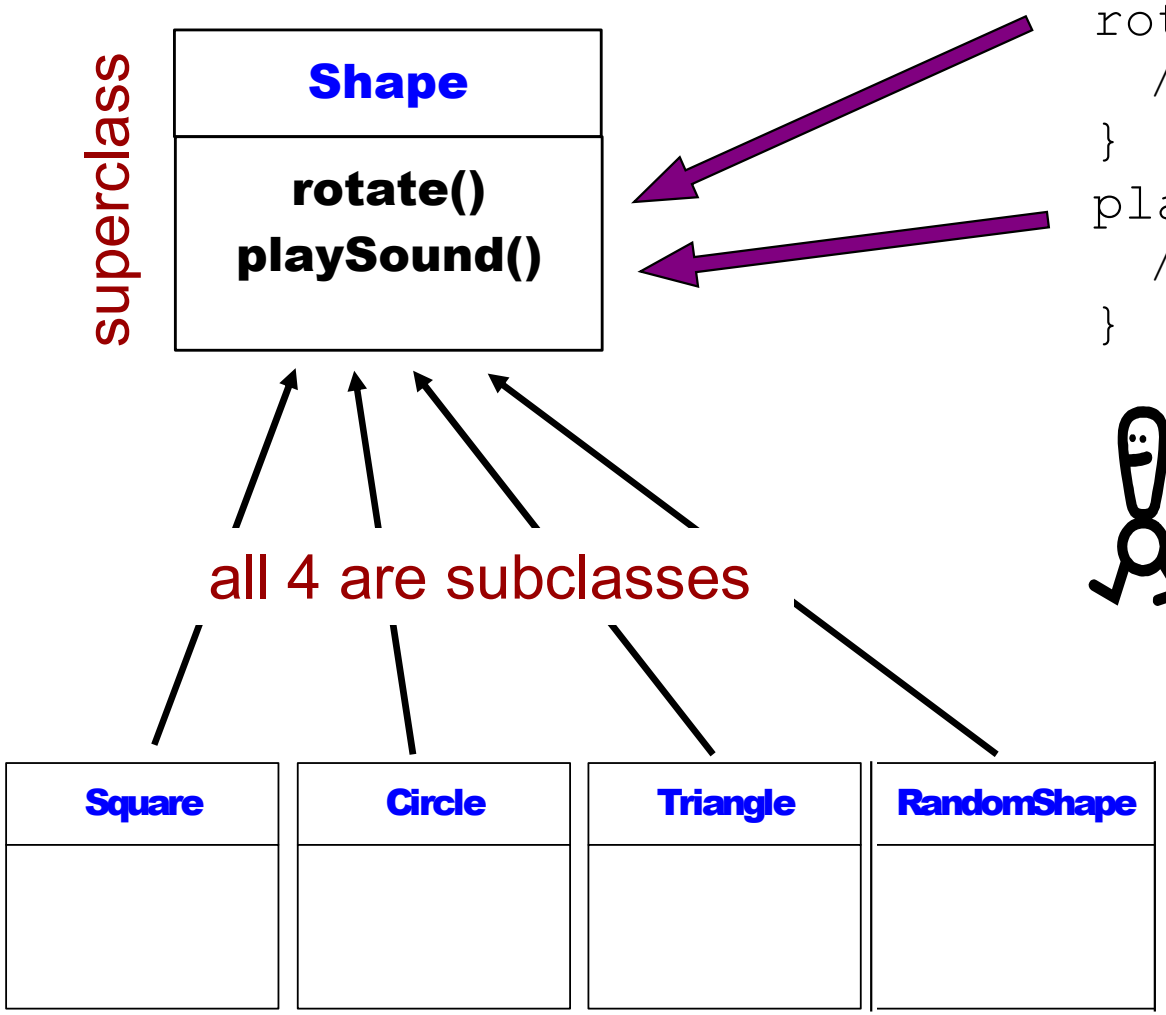


Look at what they have in common!

Abstract out the common features



Inheritance




```
rotate() {
  // generic code to rotate
}
playSound() {
  // generic code to play sound
}
```

If the **Shape** class has the functionality, then all subclasses or child classes do too!

All these shapes get the generic methods without any additional code!

"**Square** inherits from **Shape**"

But how does this help?

Specialising

- In the case of the **`RandomShape`**, we provide our own “random shape” specialisations!

```
RandomShape
int xPoint
int yPoint
rotate() {
  // code to rotate
}
playSound() {
  // code to play
  // sound for RandomShape
}
```

Method Overriding – we redefine the inherited methods!

Every object has its own behaviour!

… and things for you to try out!